Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Next-Generation Tomographic Instrumentation for Nuclear Safeguards Verification			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy and Resources			
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	*Radiation Management Technologies* *Nuclear Plant Instrumentation and Control*		
Concept of Proposed Technology	*The **spectroscopy-capable Gamma Emission Tomography (SGET) system** that we propose to develop in collaboration with SMEs enables flexible, deployment-ready, and nondestructive **verification of spent fuel assemblies across the nuclear fuel cycle, including reactor sites, storage facilities, and reprocessing plants**. By combining gamma-ray spectroscopy with tomographic imaging and advanced reconstruction algorithms, SGET reconstructs three-dimensional, rod-level gamma activity within an assembly, identifying fuel-rod configuration, inferring burnup characteristics, and estimating inventories of safeguards-relevant nuclides. These capabilities support high-confidence confirmation of fuel integrity and detection of anomalies such as missing, substituted, or damaged rods, as well as partial defect scenarios. At reactor sites, SGET provides independent verification prior to fuel transfer; in interim and long-term storage, it enables periodic re-verification without intrusive handling; and at reprocessing facilities, it strengthens front-end material control by reconciling declared and measured inventories before bulk processing. Designed to be*			

	cost-effective and field deployable, SGET reduces inspection burden while increasing measurement fidelity, supports safeguards-by-design approaches, and delivers transparent, verifiable data products that enhance material control and accountability and strengthen confidence in spent fuel management.
Rationale for Joint R&D Project	*This joint R&D project addresses a shared need to strengthen nondestructive verification and safeguards capabilities for spent nuclear fuel in an increasingly complex advanced and nuclear fuel cycle. Small and medium-sized enterprises in the Republic of Korea provide technical support to its nuclear energy and nonproliferation regime. Compliance with these requirements has driven Korea to develop advanced practices in material accountability, transparency, and regulatory implementation, making it a critical source of operational and safeguards insight [1-3].* *At the same time, our group at the University of Illinois Urbana-Champaign in the United States brings long-standing expertise in radiation detection, imaging, and safeguards technologies for nonproliferation applications, including advanced instrumentation, algorithms, and field-deployable verification systems [4-6]. By combining Korea's operational experience under stringent safeguards obligations with UIUC technical leadership in radiation detection and nondestructive assay, this collaboration enables solutions that are both technically advanced and practically deployable.* *Joint development ensures that the resulting technology is responsive to real-world regulatory and operational constraints while advancing state-of-the-art measurement performance. This partnership is therefore mutually beneficial,* ***strengthening domestic and international safeguards capabilities in both countries while contributing to global nonproliferation and nuclear security objectives****.*
Keywords	1. Gamma-ray passive tomography 2. Spent nuclear fuel 3. Nuclear safeguards 4. Instrumentation and control 5. Nuclear power plants

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)

R&D Objectives

The Nuclear Measurement lab (NML) at UIUC, led by Dr. Di Fulvio and Dr. Fang, design, simulate, develop, and characterize radiation detection systems, with a focus on neutron spectroscopy and imaging, and use Bayesian and deep learning algorithms to maximize the information retrieved from such systems The research in NML has applications in the areas of nuclear safeguards, nuclear nonproliferation, and homeland security. Examples relevant to this proposal include the development of novel fast neutron multiplicity counters (FNMC) and tomography-capable multiplicity counters, for the characterization of subcritical assemblies and spent fuel, respectively and tested such devices on special nuclear materials at US national laboratories. This research has had a direct impact on the field: the International Atomic Energy Agency is currently adopting FNMCs for the uranium assay in fresh fuel. Among other efforts, Dr. Di Fulvio is the UIUC lead PI of the Nuclear Science and Security Consortium, sponsored by the National Nuclear Security Administration.

Within this framework, the group has actively contributed to work related to passive gamma emission tomography (PGET), which is the focus of this proposed effort.

The PGET has been authorized by the IAEA for verification of spent nuclear fuel stored in water pools. As shown in Fig. 2a, the system revolves around the spent fuel assembly (SFA) and records gamma-ray counts at different angles, forming a sinogram (Fig. 1b). By applying appropriate tomographic techniques, a two-dimensional (2D) cross-sectional image of the SFA is obtained (Fig. 1c). Automated fuel pin identification is then performed to detect removed or replaced fuel pins (Fig. 1d), and their activities are quantified (Fig. 1g).

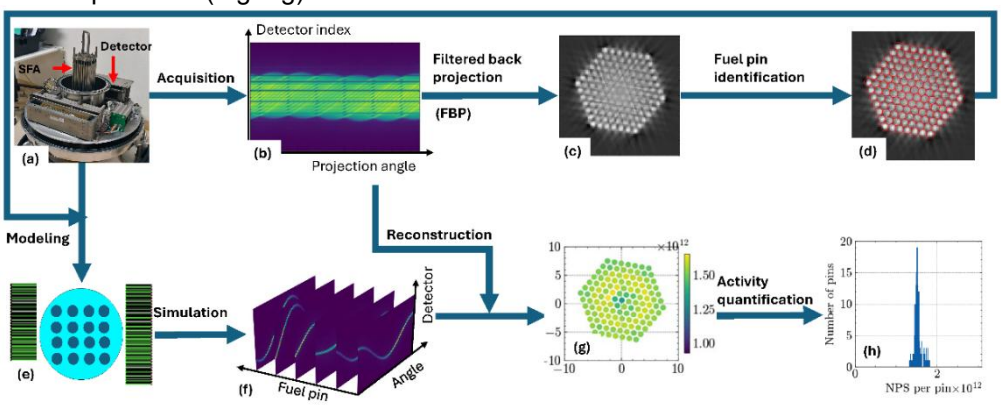

Figure 1. PGET inspection of an SFA. (a) PGET system with a mock-up fuel assembly inside. (b) Detector counts at different angles known as the sinogram. (c) A 2D image reconstructed from the sinogram. (d) Identified fuel pins. (e) A linear forward model of PGET. (f) The collection of simulated sinograms in response to each individual fuel pin, known as the system response matrix. (g) Distribution of number of photons emitted per pin, proportional to the fuel pin activity. (h) Histogram of pin activities.

In our prior work, we developed a suite of advanced algorithms addressing all three tasks on both simulated and experimental data. As shown in Fig. 2a–c, for three different VVER-1000 assemblies, the algorithms correctly identified 100% of removed rods [4]. Figures 2d–e further demonstrate that the method achieves absolute activity quantification in simulation studies [5], representing, to the best of our knowledge, the only approach with this capability. The methodology was also applied to measurements of VVER-440 assemblies, with all fuel pins clearly resolved and the central water channel successfully detected [5]. Overall, this approach is well suited for practical inspection tasks and provides a strong basis for the proposed work.

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While the current PGET system has successfully demonstrated rod-level verification, it suffers from several practical limitations. These include: (i) high system cost, driven by the need for a large number of detectors and mechanically complex collimator assemblies; (ii) limited spectral flexibility, as high data throughput and analog electronics prevent storage of full energy spectra from individual detectors during tomographic operation, forcing reliance on fixed energy windows that cannot be adjusted after acquisition; and (iii) non-uniform detector gain without a robust compensation method, which introduces stripe artifacts in the sinograms (Fig. 2b) and significantly complicates quantitative activity reconstruction.

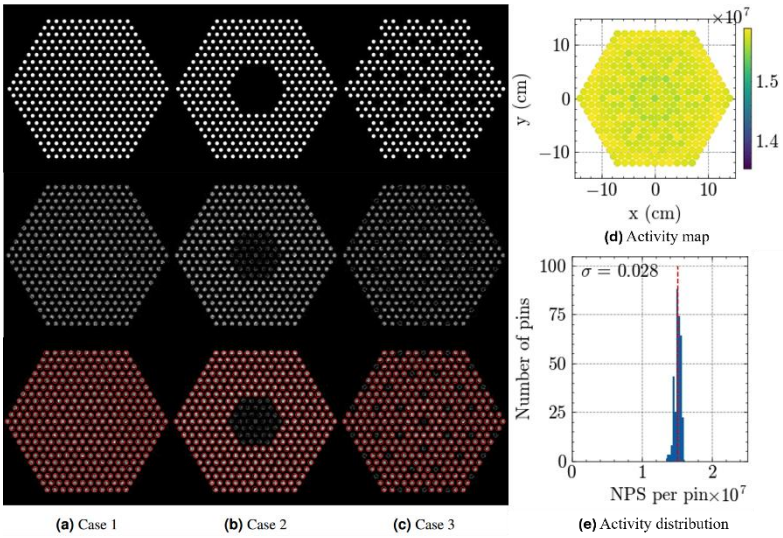

Figure 2. PGET inspection results. (a-c): Ground truth (top), reconstructed image (mid), identified fuel pins (bottom) [5]. Case 1: Full assembly. Case 2: Central rods missing. Case 3: Surrounding rods missing. (d): Reconstructed pin-level activity and (e): activity distribution with ground truth shown as red line [6].

We propose to build a cost-effective spectroscopy-capable PGET system that encompasses a high-resolution gamma-ray detector, a parallel-hole tungsten collimator, and a set of translation and rotation stages for detector movement. Compared to the traditional PGET system, the detector-collimator assembly is substantially simplified, resulting in significant cost reduction while enabling full spectroscopic data acquisition.
Importantly, the use of a single detector fundamentally eliminates inter-detector gain variation and the associated need for gain correction. By removing a dominant source of systematic error, our design is expected to substantially improve reconstruction quality and the accuracy of pin activity quantification.

R&D Target Specifications

This R&D proposed project targets a spectroscopy-capable Gamma Emission Tomography system that delivers rod-level verification performance comparable to existing PGET systems while significantly reducing cost and system complexity. Current industrial PGET systems cost approximately $0.5–1M, largely due to large CZT detector arrays and complex collimation. The proposed design targets a >5× reduction in detector count by replacing fixed arrays with a single high-resolution gamma detector mounted on combined translation and 360° rotation stages, while maintaining rod-level spatial resolution.

The system is designed to complete a full spent fuel assembly assay within ≤1 hour, consistent with item-verification time constraints. Spatial resolution targets are ≈10–15 mm, sufficient to discriminate individual fuel rods, enabled through optimized collimator geometry and detector dimensions. Full energy spectra (0–3 MeV) will be recorded at every scan position, allowing post-acquisition energy-window selection and simultaneous use of multiple gamma-ray peaks to improve reconstruction quality and reduce required scan positions.

Energy resolution targets are ≤2–3% at 662 keV for CZT (or equivalent performance with LaBr3:Ce or HPGe), with optional Compton suppression to enhance peak-to-Compton ratios. Software performance targets include automated pin identification, absolute activity quantification, and rod-level burnup estimation using both deterministic and machine-learning-based reconstruction methods.
The feasibility of this approach, particularly on the computational side, is provided by our extensive experience in simulating PGET response and processing simulated and measured data [4-6]. In our most recent work [4] we have developed an advanced physics-based reconstruction framework that enables near-real-time, rod-level activity reconstruction despite strong self-shielding and scattering effects. The approach formulates image reconstruction as a linear inverse problem using a forward model that explicitly accounts for collimator septal penetration and detector scattering, while exploiting full spectroscopic information. The resulting forward model enables rapid sinogram simulation and

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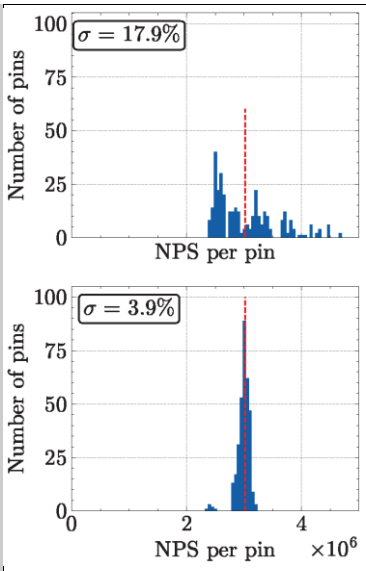

system matrix generation at speeds more than **10^5 times faster than 3D Monte Carlo methods**, while maintaining high fidelity. Validation against MCNP simulations for VVER-1000 and VVER-440 spent fuel assemblies shows agreement within **3.7%** in total counts. Using this framework, we achieved **100% fuel-pin identification** and an average **2.3% uncertainty in activity quantification** in simulation, and successfully reconstructed all pins including innermost rods and the water channel, from measured VVER-440 data. Figure 3 shows an example of improvement in activity estimate of our method, when compared to traditional filtered backprojection (FBP) with attenuation correction. This combination of accuracy, computational efficiency, and spectroscopic capability provides a robust and validated foundation for the proposed work and for deployment-relevant algorithm development.

Figure 3. Comparison of histograms of reconstructed fuel pin activities. (Top) FBP with attenuation correction. (Bottom) Reconstruction using the inverse approach. The red line represents the ground truth. σ is the relative standard deviation of the activity distribution with respect to the mean [4].

R&D Plan

The objective of this effort is to design, build, and demonstrate a cost-effective spectroscopy capable Gamma Emission Tomography system for nondestructive spent fuel verification. The research approach integrates hardware development, data acquisition, and advanced data analysis, with progressive validation through simulation and experimental measurements.

Task 1: Hardware Design and Integration

The UIUC team will design and assemble the Gamma Emission Tomography hardware, consisting of a high-resolution gamma-ray detector, a parallel-hole tungsten collimator, and motorized translation and rotation stages. The system will enable 360° scanning of a spent fuel assembly using a single detector to eliminate inter-detector gain variation. Detector selection (CZT baseline, with $LaBr_3$:Ce or HPGe as alternatives) and collimator geometry will be optimized to balance detection efficiency and rod-level spatial resolution. Modular mechanical design will allow detector replacement and configuration changes for different fuel types.

Milestone 1.1: Mechanical design of collimator and motion stages.

Milestone1.2: Fully integrated SGET hardware platform capable of controlled translation and rotation.

Task 2: Data Acquisition and Control Software

A graphical user interface will be developed to control detector motion, manage scan parameters, display sinograms, and store raw spectral data. The system will record full energy spectra at all angular and translational positions, enabling post-acquisition selection of energy windows.

Milestone 2.1: Data acquisition architecture and interface design.

Milestone2.2 : Operational data acquisition software with full spectral data storage.

Task 3: Physics-Based Reconstruction and Analysis

Deterministic reconstruction algorithms based on a physics-based forward model will be implemented to account for collimator penetration and detector scattering. These methods will be used for pin identification, activity quantification, and fuel integrity assessment. Energy-resolved sinograms will be used to estimate rod-level burnup using $^{134}Cs/^{137}Cs$ ratios.

Milestone 3.1: Deterministic reconstruction and pin identification

Milestone 3.3: Validated reconstruction and activity quantification using simulated and measured data.

Task 4: Machine-Learning-Based Analysis

In parallel, end-to-end machine-learning models will be developed to infer fuel configuration and burnup directly from raw sinograms and spectral data. Results will be compared with deterministic

methods.
Milestone 4.1: ML model development and training.
Milestone 4.2: Demonstration of ML-based reconstruction and burnup estimation.

Task 5: Experimental Demonstration and Evaluation
The completed SGET system will be tested using spent-fuel-like materials to evaluate imaging performance, assay time, and robustness. Results will be compared against prior PGET data where available.
Milestone 5.1: Experimental demonstration of rod-level verification within a one-hour assay time.

The table below outlines a progression of the tasks and milestone sequence.

Timeline (quarters (Q) and milestones (M))	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Task 1 - Hardware Design and Integration			M1.1						M1.2			
Task 2 - Data Acquisition and Control Software				M2.1			M2.2					
Task 3 - Physics-Based Reconstruction and Analysis		M3.1				M3.2						
Task 4 - Machine-Learning-Based Analysis							M 4.1			M 4.2		
Task 5 - Experimental Demonstration and Evaluation												M5

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The UIUC team, led by Dr. Angela Di Fulvio (PI) and Dr. Ming Fang (PI), will be responsible for the core technical development of the spectroscopy-capable Gamma Emission Tomography system. UIUC will lead the design and configuration of the data acquisition architecture, development of physics-based forward models, and implementation of advanced deterministic and machine-learning reconstruction algorithms for rod-level identification, activity quantification, and burnup assessment. UIUC will also oversee system integration, algorithm validation using simulated and experimental data, and overall project coordination. Required resources, including radiation detection instrumentation, computing infrastructure, and laboratory facilities for algorithm development and system testing, will be provided by UIUC.
Expected R&R of Korean SME Partner	The SME Korean team will contribute expertise in gamma-ray detection systems and spent-fuel inspection technologies, with particular emphasis on practical deployment considerations under strict safeguards and regulatory constraints. The Korean team will provide technical input on detector selection, mechanical design considerations, and operational requirements informed by Korea's extensive experience in safeguards implementation. They will also support performance evaluation, provide feedback on hardware and software design from an end-user perspective, and participate in joint testing and data interpretation activities.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The main deliverable of this project is the development of a novel NDA system that will equip the relevant stakeholders with an advanced tool for spent fuel analysis and verification. The expected outcome includes the design and characterization of the system in the laboratory (FY2027). The method will be extensively tested in simulation using state-of-the-art Monte Carlo codes and tested (FY2028-FY2029). Milestone reports will be prepared throughout the project to convey the details of the experimental setup and report findings from various laboratory experiments. Several avenues for commercialization will be explored, including patenting, licensing, and creating a startup for production and distribution.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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